

UNI-LADiR: LATENT DIFFUSION UNIFIES MULTIMODAL REASONING

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ABSTRACT

Multimodal reasoning requires models to draw on information from multiple modalities throughout the reasoning process. Yet existing methods often concatenate or interleave modality-specific thought tokens in a single sequence, leaving the model to bridge representational differences as it reasons across modalities. We introduce Uni-LaDiR (**Unified Latent Diffusion Reasoner**), a framework that brings these thoughts into a shared latent space for reasoning. A unified encoder maps teacher reasoning steps from different modalities into shared thought tokens, trained to preserve the information needed for later reasoning steps and the final answer or action. Because the same context can support multiple valid next steps, we use diffusion to predict the next block of thought tokens from the input and preceding blocks. Jointly training the encoder and diffusion reasoner with shared model weights encourages thought tokens to be both useful for the task and predictable from the available context. At inference, the model generates these tokens without teacher observations. Across eleven vision-language model (VLM) benchmarks and two vision-language-action (VLA) suites, Uni-LaDiR achieves relative gains over the strongest evaluated baselines of **7.3%** in average mathematical and logical reasoning accuracy and **6.1%** in RL Bench manipulation success.

1 INTRODUCTION

A world can be observed in different modalities. An image shows how it looks like, language describes what is happening, and 3D point clouds reveal the geometry and robot states represent the interaction. These modalities provide complementary evidence. Many multimodal problems can only be solved by combining them (Lu et al., 2022b; 2024; Hao et al., 2025) and a capable multimodal reasoner should therefore be able to use information from different modalities in its reasoning process (Gao et al., 2024; Gu et al., 2025).

Recent work has explored several ways to combine multimodal information in Chain-of-Thought reasoning. For vision-language models (VLMs), prior work interleaves text with visual reasoning steps, where the visual content takes the form of generated images (Li et al., 2025c; Gu et al., 2025) or latent visual tokens (Li et al., 2025b; Yang et al., 2025). For vision-language-action models (VLAs), prior work combines language, visual inputs, and actions (Brohan et al., 2023; Kim et al., 2024; Zhao et al., 2025), with some methods also incorporating 3D information and robot states (Zhen et al., 2024; Qu et al., 2025; Liu et al., 2026). Despite their differences, most methods follow a similar pattern: each modality is encoded into its own tokens or features, which are then concatenated or interleaved in the LLM sequence (Gao et al., 2024; Li et al., 2025b; Yang et al., 2025; Liu et al., 2026). As a result, reasoning still happens over modality-specific representations. When intermediate reasoning repeatedly transfers information between modalities, this entangles *cross-modal representation conversion* with *reasoning dynamics*. The model needs to determine not only how the underlying reasoning state should change, but also how that state is expressed in the representation of the next modality.

We argue that *while perception can remain modality-dependent, reasoning dynamics should operate over a unified reasoning space*. Each modality is specialized for extracting different evidence from the world, but reasoning only requires the task-relevant information that this evidence implies. Once such information is encoded into a shared space, subsequent reasoning can evolve independently of

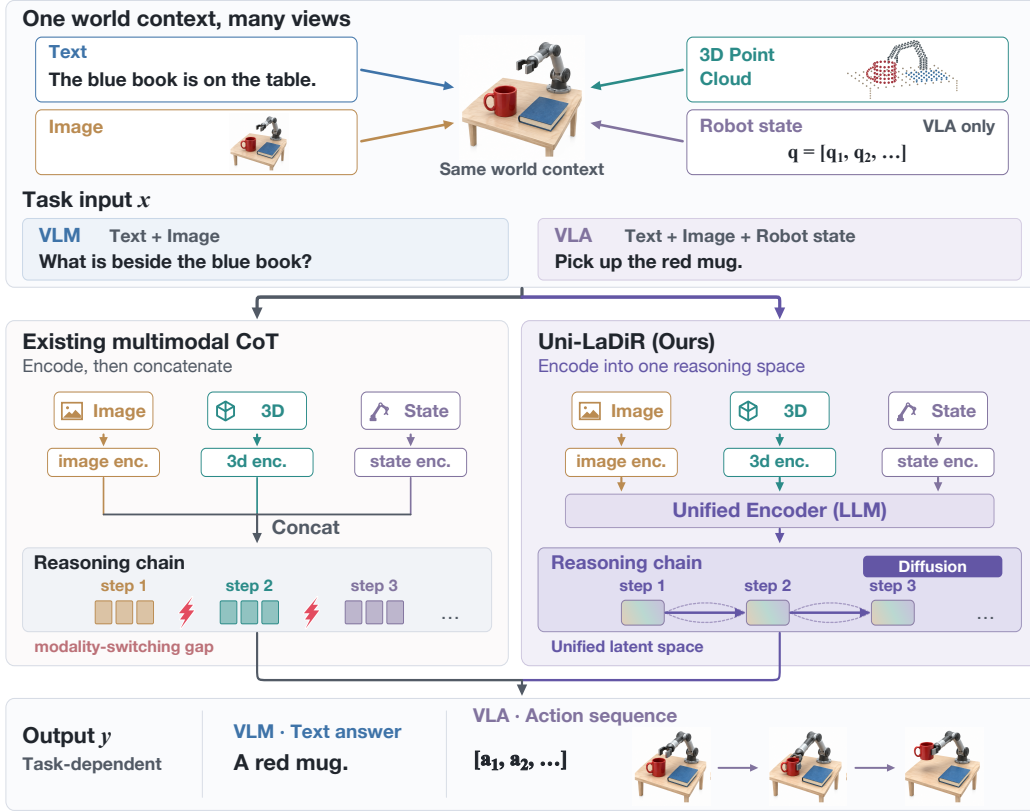


Figure 1: **Multimodal reasoning in a shared latent space.** Modality-specific reasoning (left) switches representations between steps. Uni-LaDiR (right) uses a shared latent space and diffusion to generate thought tokens for answers or actions.

the observation modality. This design gives subsequent reasoning a shared token interface instead of requiring modality-specific latent formats (Kaelbling et al., 1998; Hafner et al., 2019; 2020).

Based on this idea, we introduce Uni-LaDiR, a unified latent diffusion framework for multimodal reasoning. Unlike prior approaches that directly concatenate outputs from modality-specific encoders, Uni-LaDiR introduces a unified encoder that maps heterogeneous modality features into a *shared latent reasoning space*. Specifically, every teacher reasoning step is encoded into a fixed-width latent block as thought tokens, and each block is trained to support the generation of future reasoning steps and the final task output, rather than to reconstruct its source modality. This encourages the thought tokens to preserve information according to its utility for subsequent reasoning, while factoring out modality-specific representational differences that are irrelevant to the task.

However, learning to predict these thought tokens is not trivial. First, the next reasoning step is often not unique: the same context may support multiple valid intermediate thoughts or actions. We therefore use diffusion to model the conditional distribution of the next thought tokens given the input and preceding thought tokens (Ho et al., 2020; Rombach et al., 2022; Chi et al., 2023; Kang et al., 2025; 2026). Second, useful thought tokens should satisfy two properties: *reasoning sufficiency* and *context predictability*. They should preserve the information needed for subsequent reasoning and the final answer, while remaining predictable from the available context at inference time. We therefore jointly train the unified encoder and the latent diffusion reasoner, allowing the representation space and the reasoning model to adapt to each other. This encourages thought tokens that retain task-relevant information and can be generated from the available context.

We evaluate Uni-LaDiR on eleven VLM reasoning benchmarks and two VLA manipulation suites. Compared with the strongest reported modality-specific latent reasoning baseline in each setting, Uni-LaDiR improves mean accuracy by **7.3%** relative across four mathematical and logical VLM benchmarks and mean success by **6.1%** relative across ten RL-Bench manipulation tasks. Within the

unified latent space, flow matching improves the average by **14.9%** over direct L2 prediction and **19.5%** over cosine similarity loss (Figure 6). These results support Uni-LaDiR’s approach of separating modality-specific perception from shared latent reasoning, with thought-token representations and their prediction learned together for the task.

2 PRELIMINARIES

Problem formulation. Given an input x containing multimodal observations and a query or instruction, the reasoner generates K intermediate thought-token blocks $z_{1:K}$ before predicting the task output y , which is a text answer for VLMs or a robot action for VLAs. The intermediate blocks integrate task-relevant evidence across modalities and mediate prediction of y .

Latent multimodal CoT. Existing latent multimodal CoT represents intermediate reasoning with modality-specific latent tokens. In VLMs, these tokens can encode intermediate visual content (Li et al., 2025b; Yang et al., 2025); in VLAs, they can encode future visual observations, 3D geometry, or robot states (Liu et al., 2026). The resulting blocks are concatenated or interleaved in the model sequence:

$$p_\theta(z_{1:K}, y \mid x) = \underbrace{\prod_{i=1}^K p_\theta(z_i \mid [x; z_{<i}])}_{\text{intermediate reasoning}} \underbrace{p_\theta(y \mid [x; z_{1:K}])}_{\text{final output}}. \quad (1)$$

Here $[\cdot; \cdot]$ denotes sequence concatenation. During training, a modality-specific encoder maps the i -th teacher reasoning step, such as an intermediate image or robot state, to features s_i that supervise the corresponding thought block z_i . At inference, the teacher steps are unavailable, and the model must generate each z_i from x and the preceding thought blocks.

3 UNIFIED LATENT DIFFUSION REASONING

Uni-LaDiR unifies different reasoning steps from different modalities into the same latent space to provide a shared interface for reasoning across modalities. In this section, we first formalize the requirements of this shared reasoning space (Section 3.1), then describe how teacher steps are mapped to shared thought tokens (Section 3.2), how downstream supervision grounds these tokens in subsequent reasoning and task prediction (Section 3.3), and how latent diffusion learns to generate them from the context available at inference (Section 3.4). Figure 2 provides an overview.

3.1 MOTIVATION

Modality-specific latent CoT couples each reasoning step with a change in representation: predicting the next step requires determining not only its task-relevant content, but also how that content is expressed in the target modality. We instead require every intermediate step to inhabit a shared latent reasoning space, independent of the modality from which its supervision originates.

A useful shared representation must satisfy two complementary properties. *Reasoning sufficiency* requires each thought block to retain the information needed for subsequent reasoning steps and the final output. *Context predictability* requires the same block to be recoverable from the input and preceding thought tokens available at inference. We enforce the former through downstream grounding supervision and the latter through latent diffusion joint training, while sharing the backbone between representation construction and generation (Duggal et al., 2026).

3.2 CONSTRUCTING SHARED THOUGHT TOKENS

We train a unified encoder to map each teacher reasoning step to the corresponding thought tokens z_i^* in a unified reasoning space (Figure 2a). After modality-specific encoding, we append a fixed-length block of learnable embeddings q to each teacher feature s_i . Then our unified encoder maps the last hidden states at the q positions to the clean thought tokens z_i^* .

Every modality uses the same q , output projection, and number of thought tokens per block. During training, we pack multiple teacher steps and their corresponding copies of q into one forward pass. The construction mask isolates the steps: each copy of q attends only to itself and its associated teacher features, preventing information exchange across reasoning steps (Figure 2a). Thus, all teacher modalities are now encoded into a shared latent space, where downstream supervision determines which information is retained for better subsequent reasoning.

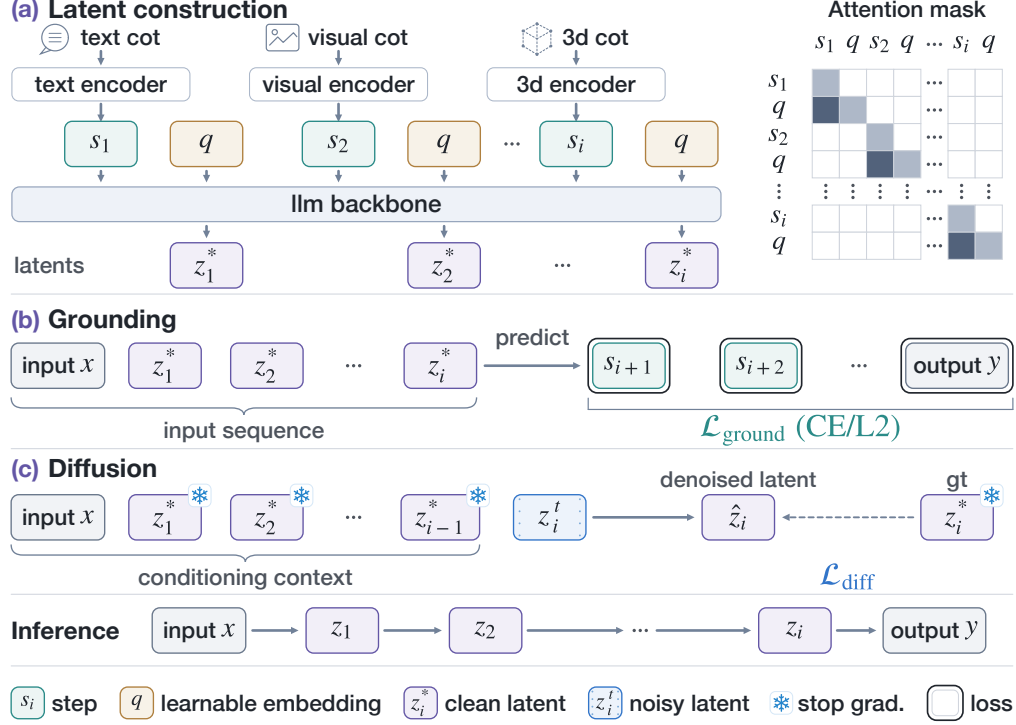


Figure 2: **Uni-LaDiR overview.** During training, a shared backbone (a) encodes teacher steps into thought tokens, (b) grounds them through downstream predictions, and (c) learns their generation by diffusion. At inference, it generates thought tokens from the task input to predict an answer or action.

3.3 GROUNDING THOUGHT TOKENS IN DOWNSTREAM REASONING

Grounding trains each thought block to preserve information needed by subsequent reasoning. Given x and the prefix $z_{<i}^*$, the backbone predicts the next teacher target s_i^{GT} ; given the complete sequence $z_{1:K}^*$, it predicts the final output y (Figure 2b). For example, z_1^* contributes to predicting s_2^{GT} , while (z_1^*, z_2^*) contribute to predicting s_3^{GT} . A grounding attention mask computes these predictions in one forward pass while exposing each prediction only to its permitted thought-token prefix and hiding later blocks and raw teacher steps (Appendix A.1).

Let $\hat{s}_i$ and $\hat{y}$ denote the corresponding predictions. We optimize

$$\mathcal{L}_{\text{ground}} = \underbrace{\sum_{i=2}^K \ell(\hat{s}_i, s_i^{\text{GT}})}_{\text{reasoning steps}} + \lambda_y \underbrace{\ell(\hat{y}, y)}_{\text{final output}}, \quad \ell(\hat{u}, u) = \begin{cases} \text{CE}(\hat{u}, u), & u \text{ is text,} \\ \|\hat{u} - u\|_2^2, & u \text{ is continuous.} \end{cases} \quad (2)$$

Here u denotes either a reasoning-step target or the final output, and λ_y weights final-output supervision. The continuous reasoning-step targets are image and 3D point-cloud features and robot-state vectors, supervised by the L2 term above. For VLAs, the final action y is instead produced by a flow-matching action head, so its term $\ell(\hat{y}, y)$ is that head’s flow-matching loss rather than an L2 regression. The targets s_i^{GT} and y are fixed, and any encoder used to construct target features is frozen. Gradients from $\mathcal{L}_{\text{ground}}$ pass through z_i^* into the unified encoder and shared backbone, learning the latent thought tokens to optimize their utility for subsequent reasoning.

3.4 GENERATING THOUGHT TOKENS WITH JOINT DIFFUSION TRAINING

The diffusion objective trains the backbone to generate each thought block from the context available at inference (Figure 2c). Diffusion models a conditional distribution over the next thought tokens, allowing different valid thoughts under the same input and preceding thought tokens (Kang et al., 2025; 2026). During training, Gaussian noise perturbs each clean block z_i^* to produce z_i^t . Conditioned on x , the preceding clean blocks $z_{<i}^*$, and noise level t , the shared backbone predicts the denoising velocity $\hat{v}_i$.

We again process all reasoning steps in one forward pass. The diffusion mask restricts each noisy block to its own tokens, x , and the preceding clean thought blocks, preventing access to future thought blocks (Appendix A.1). We optimize the flow-matching objective (Lipman et al., 2023)

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t,\epsilon} \left[\sum_{i=1}^K \|\hat{v}_i - v_i^*\|_2^2 \right], \quad (3)$$

where v_i^* is the target velocity induced by the sampled noise ϵ and time t . The predicted velocity guides denoising to generate the thought-token block $\hat{z}_i$. Here, t indexes denoising within a block, while i indexes reasoning steps.

Joint training. Inspired by Duggal et al. (2026), we jointly train the unified encoder and diffusion reasoner with a shared backbone, optimizing the latent thought tokens to be both useful for downstream reasoning and predictable from the available context. We optimize

$$\mathcal{L} = \mathcal{L}_{\text{ground}} + \lambda \mathcal{L}_{\text{diff}}, \quad (4)$$

where λ balances the two objectives. On the diffusion branch, we stop gradients through the clean thought tokens z_i^* , including both the conditioning prefix and the noising target. Thus, $\mathcal{L}_{\text{ground}}$ updates the unified encoder and shared backbone through z_i^* , whereas $\mathcal{L}_{\text{diff}}$ updates the shared backbone while treating the current clean tokens as fixed targets. Because the same backbone constructs and predicts the latent representations, the two objectives jointly improve the thought-token representations and the model that predicts them.

Inference. At inference, the teacher reasoning steps and unified encoding path are not required. Starting from x , the model generates $z_1, \dots, z_K$ sequentially by denoising one latent block at a time, with each block conditioned on the previously generated blocks (Figure 2, bottom). The complete latent sequence then conditions prediction of the final answer or action y .

4 EXPERIMENTS

Our experiments test whether a shared latent reasoning space improves both visual reasoning and robotic manipulation. Section 4.1 introduces the benchmarks and baselines; Section 4.2 reports task performance and manipulation inference efficiency; Section 4.3 isolates the effect of unifying teacher modalities; and Section 4.4 examines the training objectives and joint optimization.

4.1 EVALUATION PROTOCOL

Visual reasoning. We train on Zebra-CoT (Li et al., 2025a), which interleaves text and visual reasoning steps, and use Qwen2.5-VL-7B for the main experiment. We report accuracy on seven vision-centric benchmarks: VisualPuzzles (Song et al., 2025), ChartQA (Masry et al., 2022), V* (Wu & Xie, 2024), BLINK-Jigsaw (Fu et al., 2024), MMVP (Tong et al., 2024b), SAT (Ray et al., 2024), and CV-Bench (Tong et al., 2024a). We also evaluate four mathematical or logical benchmarks: MathVista (Lu et al., 2024), MathVision (Wang et al., 2024b), VisuLogic (Xu et al., 2025b), and EMMA (Hao et al., 2025). Appendix B.3 reports results with additional backbones.

VLA manipulation. We evaluate LIBERO (Liu et al., 2023) and RLBench (James et al., 2020). In this setting, future images, 3D point clouds, and robot states serve as training-only teacher CoTs; continuous actions remain task outputs. Following Liu et al. (2026), our VLA backbone is a mixture-of-transformers (MoT) that couples two Janus-Pro-1.5B experts (3.3B parameters total), and actions are produced by a flow-matching action head trained with the same action objective. We report success rate and, on RLBench, control frequency without action chunking.

Baselines. For VLMs, we compare with the Qwen2.5-VL base model and explicit reasoners DeepEyes (Zheng et al., 2025), PixelReasoner (Wang et al., 2025), Vision-R1 (Huang et al., 2025b), and Open-Vision-Reasoner (Wei et al., 2025). Latent reasoning baselines include LVR (Li et al., 2025b), Mirage (Yang et al., 2025), CoVT (Qin et al., 2025), Monet (Wang et al., 2026), VaLR-M (Jeon et al., 2026), SLVR (Xu et al., 2026a), Mull-Tokens (Ray et al., 2025), and ILVR (Dong et al., 2026). For VLAs, we consider three groups. *Action policies* include OpenVLA (Kim et al., 2024), SpatialVLA (Qu et al., 2025), CogACT (Li et al., 2024), $\pi_{0.5}$ (Physical Intelligence et al., 2025), OpenVLA-OFT (Kim et al., 2025), HybridVLA (Liu et al., 2025), and UniVLA (Bu et al., 2025). *Explicit reasoning models* include ManipLLM (Li et al., 2023a),

Table 1: **Comparison on VLM reasoning benchmarks.** Accuracy (%) of Uni-LaDiR and baselines with Qwen2.5-VL-7B on (a) vision-centric and (b) mathematical and logical benchmarks. Best results are highlighted in bold.

(a) Vision-Centric Perception, Search, and Spatial Reasoning.								
Method	VisPuz.	ChartQA	V*	BLINK-J	MMVP	SAT	CV-B	Avg.
Explicit Text and Visual Reasoning								
Qwen2.5-VL (base)	34.75	78.12	76.44	59.33	77.33	51.33	75.20	64.64
DeepEyes	48.03	82.52	90.10	72.00	78.00	64.00	84.00	74.09
PixelReasoner	44.01	81.52	84.30	70.00	76.00	60.00	81.99	71.12
Visual Latent Reasoning								
LVR	45.98	81.00	81.70	52.00	71.70	60.00	84.00	68.05
Mirage	44.01	80.52	83.77	88.00	74.00	72.00	83.02	75.05
CoVT	41.01	79.52	78.00	68.00	58.70	61.33	80.00	66.65
Monet	35.02	79.00	83.25	65.33	50.00	55.33	71.10	62.72
VaLR-M	45.03	82.00	86.90	72.00	60.30	65.33	87.60	71.31
SLVR	34.20	77.20	82.20	68.00	74.00	62.00	83.02	68.66
Mull-Tokens	48.03	81.00	83.77	74.67	76.00	77.00	84.00	74.92
Unified Latent Reasoning								
Uni-LaDiR	57.36	84.20	83.77	75.33	82.00	65.33	86.20	76.31

(b) Visual Mathematics and Logic.					
Method	MathVista	MathVision	VisuLogic	EMMA	Avg.
Explicit Text and Visual Reasoning					
Qwen2.5-VL (base)	68.20	25.40	26.00	26.00	36.40
Vision-R1	73.50	40.99	15.20	24.30	38.50
Open-Vision-Reasoner	69.50	51.80	27.20	29.99	44.62
DeepEyes	70.10	26.60	27.80	29.02	38.38
PixelReasoner	68.90	34.21	23.40	23.50	37.50
Visual Latent Reasoning					
Mirage	68.60	40.49	26.60	27.00	40.67
Monet	62.50	33.82	10.60	22.10	32.26
ILVR	71.10	49.01	29.30	33.30	45.68
Unified Latent Reasoning					
Uni-LaDiR	73.20	54.28	33.00	35.62	49.03

WorldVLA (Cen et al., 2025), CoT-VLA (Zhao et al., 2025), FlowVLA (Zhong et al., 2025), and FUTURE-VLA (Xu et al., 2026b). *Latent reasoning models* include FiS-VLA (Chen et al., 2025a), ConsisVLA-4D (Li et al., 2026c), LaST₀ (Liu et al., 2026), AVA-VLA (Lei & Shan, 2026), and PearlVLA with and without PRL (Yang & Shan, 2026). Full settings appear in Appendix B.

4.2 MAIN RESULTS

VLM visual reasoning. On the seven vision-centric benchmarks, Uni-LaDiR improves average accuracy over Qwen2.5-VL-7B by **18.1%**; on the four mathematical and logical benchmarks, the relative gain is **34.7%** (Table 1). It also outperforms the strongest modality-specific latent baseline in each group: **1.7%** over Mirage and **7.3%** over ILVR on average. The gains over modality-specific baselines reach **30.3%** on VisualPuzzles over Mirage, and **10.8%** on MathVision and **12.6%** on VisuLogic over ILVR. The advantage over modality-specific baselines suggests that sharing reasoning representations benefits both visual reasoning and mathematical tasks.

VLA manipulation. Uni-LaDiR improves average success over the strongest prior latent policies by **0.45 percentage points** over PearlVLA + PRL on LIBERO and **6.1%** relative to LaST₀ on RL-Bench (Tables 3 and 2). On RL-Bench, it improves nine of ten tasks and matches the remaining one; the largest gain is **16.7%** on Plants. These reported means place Uni-LaDiR among the strongest policies in this comparison. Without action chunking, Uni-LaDiR runs at **16.5 Hz**, a **7.1%** higher inference rate than LaST₀ and **15.0×** the rate of explicit CoT-VLA.

4.3 ANALYSIS

In this section, we test whether progressively unifying teacher modalities reduces the modality-switching gap and improves reasoning. All variants use the same teacher modalities, reasoning backbone, step order, latent-token budget, and training schedule; we only change how teacher signals are represented. In the separate baseline, each modality bypasses the unified encoder and directly uses its modality-specific features as latent targets. In the partially shared variants, only the specified modalities pass through the same unified encoder to form shared thought tokens, while the remaining modalities keep their original modality-specific targets. Full sharing routes all available teacher modalities through the unified encoder (Appendix B.2). As shown in Figure 3a, for visual reasoning, moving text and image from separate targets into the shared space improves mean accuracy by **9.0%**

Table 2: **Comparison of Uni-LaDiR and baselines on RL Bench.** We report mean success rates (%) and inference speed (Hz) without action chunking. Var. is the variance of mean success rate across three random seeds. Best results are highlighted in bold.

Method	Size	Box	Laptop	Toilet	Sweep	Fridge	Phone	Umbrella	Frame	Wine	Plants	Mean	Var.	Hz
Direct Policies														
OpenVLA	7B	60.0	35.0	75.0	55.0	85.0	20.0	30.0	15.0	20.0	5.0	40.0	0.02	6.3
SpatialVLA	–	80.0	70.0	85.0	20.0	80.0	15.0	25.0	40.0	15.0	30.0	46.0	0.03	7.9
CogACT	7B	90.0	80.0	95.0	50.0	85.0	50.0	55.0	45.0	30.0	25.0	61.0	0.04	9.8
$\pi_{0.5}$	3B	90.0	95.0	85.0	75.0	100.0	5.0	10.0	80.0	75.0	35.0	65.0	0.04	13.8
HybridVLA	7B	85.0	95.0	100.0	90.0	100.0	50.0	50.0	70.0	50.0	50.0	74.0	0.04	6.1
Explicit Reasoning														
ManipLLM	7B	50.0	80.0	40.0	20.0	80.0	35.0	10.0	25.0	15.0	20.0	38.0	0.04	2.2
CoT-VLA	–	95.0	75.0	100.0	80.0	65.0	50.0	40.0	50.0	55.0	50.0	66.0	0.03	1.1
Latent Reasoning														
FiS-VLA	7B	100.0	100.0	95.0	55.0	90.0	50.0	50.0	70.0	55.0	20.0	69.0	0.03	21.9
LaST ₀	3.3B	95.0	95.0	100.0	80.0	85.0	75.0	75.0	70.0	85.0	60.0	82.0	0.03	15.4
Unified Latent Reasoning														
Uni-LaDiR	3.3B	100.0	100.0	100.0	85.0	90.0	80.0	80.0	75.0	90.0	70.0	87.0	0.03	16.5

Table 3: **Comparison of Uni-LaDiR and baselines on LIBERO.** We report success rates (%) across four suites. Best results are highlighted in bold.

Method	Size	Spatial	Object	Goal	Long	Mean
Direct Policies						
UniVLA	–	92.6	93.8	86.6	63.0	84.0
OpenVLA	7B	84.7	88.4	79.2	53.7	76.5
SpatialVLA	–	88.2	89.9	78.6	55.5	78.1
CogACT	7B	97.2	98.0	90.2	88.8	93.6
$\pi_{0.5}$	3B	98.8	98.2	98.0	92.4	96.9
OpenVLA-OFT	7B	97.6	98.4	97.9	94.5	97.1
Explicit Reasoning						
WorldVLA	–	85.6	89.0	82.6	59.0	79.1
CoT-VLA	–	87.5	91.6	87.6	69.0	83.9
FlowVLA	8.5B	93.2	95.0	91.6	72.6	88.1
FUTURE-VLA	4B	89.6	99.0	95.2	81.2	91.3
Latent Reasoning						
ConsisVLA-4D	7B	98.8	99.8	98.0	95.6	98.1
LaST ₀	3.3B	99.2	99.6	98.0	95.6	98.1
AVA-VLA	7B	97.8	99.4	97.8	98.1	98.3
PearlVLA	7B	99.2	99.6	98.2	96.8	98.5
PearlVLA + PRL	7B	99.4	99.8	98.4	97.2	98.7
Unified Latent Reasoning						
Uni-LaDiR	3.3B	100.0	99.0	100.0	97.6	99.15

relative. On RL Bench, sharing any two of the visual, 3D, and state modalities improves success by **1.8–3.0%**, while sharing all three yields a **6.1%** gain over separate targets and a **3.0%** gain over the best pair (Figure 3b). These growing gains with more unified modalities suggest that our method effectively reduce the modality-switching gap, allowing the reasoning model is able to focus more on evolving the reasoning state rather than adapting between modality-specific representations.

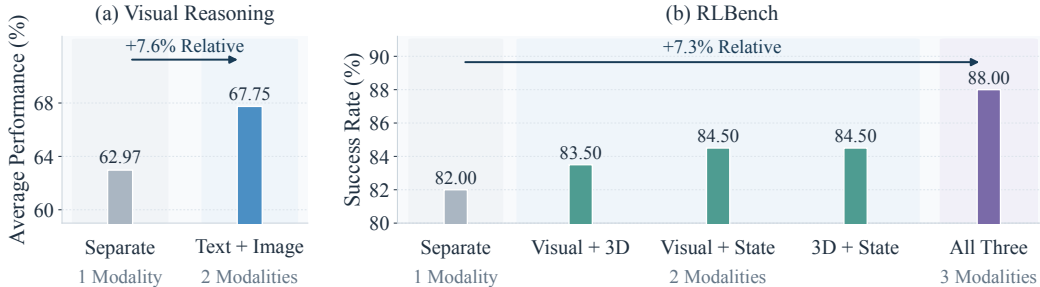


Figure 3: **Results of unifying more modalities.** (a) Qwen2.5-VL-7B mean accuracy on V*, MMVP, MathVista, and EMMA. (b) RL Bench success with different modality groupings. Error bars denote standard deviation across three seeds.

4.4 ABLATION STUDY

Grounding objective. We first test how thought tokens should be supervised. With the model, teachers, latent-token budget, and training schedule fixed, we replace source reconstruction with prediction of later reasoning steps. Both variants retain the same final-output supervision and its gradients into the encoder (Appendix B.2). Downstream supervision improves the six-benchmark average by **5.4%** relative to reconstruction, with gains of **26.0%** on V* and **7.4%** on RLBench (Figure 4). The gain spans five of six benchmarks, while MathVista decreases by 4.20 percentage points; downstream grounding improves the aggregate without benefiting every task.

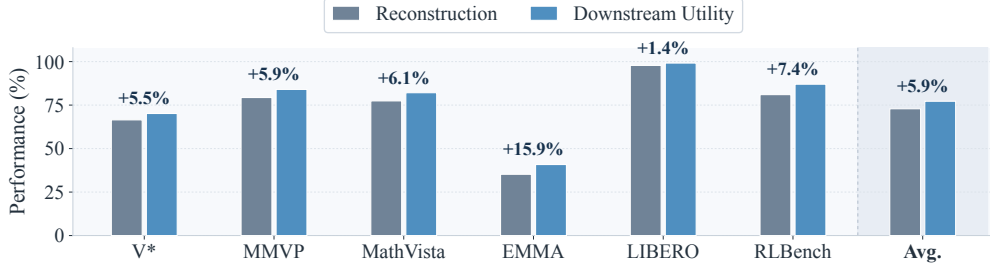


Figure 4: Results of different grounding objectives.

We compare frozen latent targets, an encoder trained before the reasoner, and joint training under the same architecture and teacher supervision. Joint training improves the six-benchmark mean by **11.3%** relative to separate stages and **21.9%** relative to frozen targets (Figure 5). Relative to separate stages, it improves all six benchmarks, with gains of **8.8%** on EMMA and **15.0%** on RLBench. These results favor learning the representation and its predictor together over fixing the representation before training the reasoner.

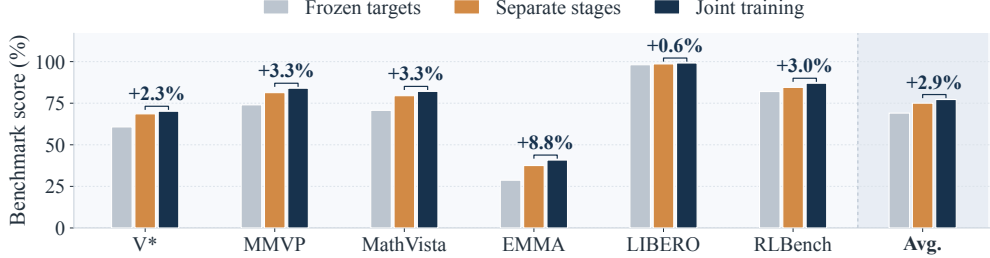


Figure 5: Results of joint versus separate training of the encoder and reasoner.

Diffusion objective. Finally, we test whether diffusion improves thought-token prediction when the latent targets are fixed. We compare flow matching with direct thought-token prediction trained using squared L2 or cosine similarity loss (Appendix B.2). Flow matching improves the six-benchmark average by **14.9%** relative to L2 and **19.5%** relative to cosine similarity loss (Figure 6). These gains support diffusion-based thought-token prediction as an effective alternative to direct regression for downstream reasoning and manipulation.

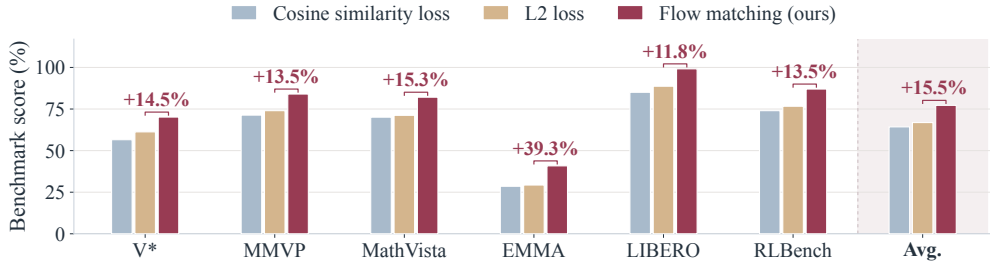


Figure 6: Results of different latent prediction objectives.

5 RELATED WORK

Unified multimodal models. Early generalist models cast heterogeneous tasks as a shared sequence problem, spanning vision–language understanding, generation, and embodied control (Wang et al., 2022; Lu et al., 2022a; 2023; Reed et al., 2022; Mizrahi et al., 2023; Bachmann et al., 2024). Large-scale vision–language models then connected pretrained language models to visual inputs through learned interfaces (Alayrac et al., 2022; Chen et al., 2022; 2023; Huang et al., 2023; Peng et al., 2023). Autoregressive any-to-any systems further represented multiple modalities in a common token stream (Sun et al., 2023a;b; Yu et al., 2023; Ge et al., 2023; Jin et al., 2023; Wu et al., 2023b; Zhan et al., 2024; Chameleon Team, 2024; Wang et al., 2024c), while hybrid autoregressive–diffusion architectures unified discrete and continuous generation (Tang et al., 2023; Zhou et al., 2024a; Xie et al., 2024; Sun et al., 2024; Wu et al., 2024; Chen et al., 2025b; Deng et al., 2025a; Xu et al., 2025a; Bai et al., 2025). Whereas these models unify what is read and generated—their inputs, outputs, and task interfaces—Uni-LaDiR unifies the reasoning in between, learning a single latent space in which the intermediate thoughts from every modality are represented and predicted.

Multimodal reasoning. Multimodal chain-of-thought methods supervise or elicit textual rationales from visual evidence (Lu et al., 2022b; Zhang et al., 2024; Rose et al., 2023; Xu et al., 2024; Dong et al., 2024; Guo et al., 2024; Yao et al., 2024; Huang et al., 2025b). A complementary line externalizes computation through programs, tools, image operations, spatial abstractions, or visual scratchpads (Gupta & Kembhavi, 2022; Suris et al., 2023; Yang et al., 2023b; Wu et al., 2023a; Hu et al., 2023; Chen et al., 2024; Hu et al., 2024; Zhou et al., 2024b). Recent work interleaves language with visual thoughts rather than forcing every step into text (Gao et al., 2024; Li et al., 2025a; Gu et al., 2025; Li et al., 2026b). In embodied settings, language plans, visual predictions, and state or action trajectories have been used to guide control (Ahn et al., 2022; Huang et al., 2022; Driess et al., 2023; Brohan et al., 2023; Jiang et al., 2022; Li et al., 2023b; Kim et al., 2024; Li et al., 2024; Zhao et al., 2025; Zhong et al., 2026; Bai et al., 2026; Liu et al., 2026). These methods still reason by switching between modality-specific representations—textual rationales, generated images, or fixed domain-specific latents—whereas Uni-LaDiR reasons inside one shared interface into which text, images, 3D structure, and robot state are all encoded; actions remain task outputs.

Latent reasoning. Latent-computation methods internalize explicit rationales, distill hidden reasoning, or allocate additional computation without decoding every step (Zelikman et al., 2022; Goyal et al., 2023; Deng et al., 2023; 2024; Pfau et al., 2024; Zelikman et al., 2024; Yu et al., 2024; Wang et al., 2024a). Recurrent and concept-level models instead treat depth, loops, or continuous semantic states as the substrate of computation (Dehghani et al., 2018; Giannou et al., 2023; Yang et al., 2023a; Barrault et al., 2024; Kohli et al., 2026; Jeddi et al., 2026; Yang et al., 2026). Continuous chain-of-thought approaches directly optimize hidden vectors that replace or compress language rationales (Hao et al., 2024; Shen et al., 2025; Xu et al., 2025c;d; Deng et al., 2025b). Multimodal variants construct continuous thoughts from visual features, dense perceptual teachers, helper images, or task loss (Pham & Ngo, 2025; Li et al., 2025b; Yang et al., 2025; Qin et al., 2025; Wang et al., 2026; Jeon et al., 2026; Viveiros et al., 2026a; Li et al., 2026a; Xu et al., 2026a; Ray et al., 2025; Hu et al., 2026), while diffusion-based approaches model multimodal or multi-step latent trajectories (Kang et al., 2025; 2026; He et al., 2023; Huang et al., 2025a; Bai et al., 2026; Liu et al., 2026; Viveiros et al., 2026b; Wu et al., 2026; Fan et al., 2026). Prior latent reasoners either compress rationales within a single modality or give each modality its own latent format; Uni-LaDiR instead learns one representation and one diffusion predictor shared across heterogeneous reasoning modalities, and directly tests whether this sharing outperforms matched modality-specific designs.

6 CONCLUSION

We introduce Uni-LaDiR, a unified latent diffusion framework for multimodal reasoning. A shared encoder maps teacher steps from different modalities into thought tokens, and a jointly trained diffusion reasoner predicts these tokens under downstream reasoning and task supervision. This gives heterogeneous intermediate steps a unified latent space, allowing the model to reason without generating modality-specific representations or accessing teacher observations at inference. Across eleven VLM benchmarks and two VLA suites, Uni-LaDiR achieves relative improvements of 7.3% and 6.1% over strong prior baselines, respectively.

AI USE STATEMENT

OpenAI Codex assisted with drafting, literature search, feedback on method formulation, mathematical objectives and experimental design, result interpretation, figures, plotting code, and numerical checks. The authors verified all reported experimental results and retain responsibility for the final text, code, citations, figures, and claims.

ETHICS STATEMENT

Experiments use existing visual benchmarks and simulated manipulation tasks. The method may inherit biases from pretrained models and teacher data; simulation performance does not establish safety or reliability on physical robots.

REPRODUCIBILITY STATEMENT

Section 3 describes the model and training objectives. Appendix A details the attention masks, and Appendix B reports evaluation protocols and supplementary results.

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A METHOD DETAILS

A.1 ATTENTION MASKS FOR PARALLEL BLOCK TRAINING

Figure 7 illustrates the three masks for K reasoning steps packed into each pass; ellipses indicate omitted intermediate blocks. The restrictions apply at every layer, preventing masked information from reaching predictions through other hidden states.

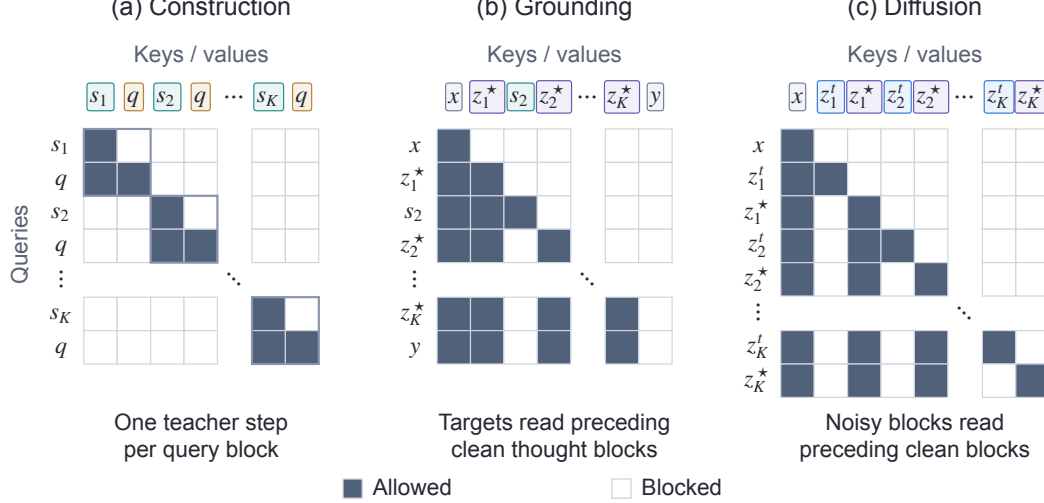


Figure 7: **Attention masks for parallel block training.** Rows attend to columns; each cell represents a token block. Grounding interleaves clean thought blocks with target streams; diffusion interleaves noisy and clean thought blocks. Clean blocks cannot read target or noisy streams.

Construction. The packed sequence contains pairs $[s_i; q]$ for all teacher steps. Attention is isolated within each pair: the learnable block reads its own teacher features and itself, but neither the task input nor other pairs.

Grounding. The packed sequence is $[x; z_1^*; s_2; z_2^*; s_3; \dots; z_K^*; y]$, where s_i and y label target prediction streams. A prediction of s_i^{GT} can read x and $z_{<i}^*$; the final-output prediction can read x and all $z_{1:K}^*$. The conditioning stream preserves this prefix order and cannot read the target streams. Different target streams are isolated from one another. Within a text target, standard causal attention permits earlier tokens of that target; continuous predictions cannot read their ground-truth values.

Diffusion. The packed sequence is $[x; z_1^t; z_1^*; z_2^t; z_2^*; \dots; z_K^t; z_K^*]$. Each noisy block z_i^t can read x , the clean prefix $z_{<i}^*$, and all tokens within itself. It cannot read its own clean target, later clean blocks, or other noisy blocks. The clean conditioning stream follows block-causal attention and cannot read noisy blocks.

B ADDITIONAL EXPERIMENTS

B.1 EXPERIMENTAL SETTINGS

Benchmarks and scoring. All methods in Tables 1 and 5 use the same evaluation examples within each benchmark. Table 4 lists the evaluation splits, sample counts, and scoring units for the eleven visual benchmarks (Song et al., 2025; Masry et al., 2022; Wu & Xie, 2024; Fu et al., 2024; Tong et al., 2024b; Ray et al., 2024; Tong et al., 2024a; Lu et al., 2024; Wang et al., 2024b; Xu et al., 2025b; Hao et al., 2025). VisualPuzzles and VisuLogic use their released evaluation collections, even when the hosting service names the split `train`. VisuLogic evaluation uses the original 1,000-question benchmark (Xu et al., 2025b), rather than the current 1,003-row hosted collection.

Table 4: **Shared visual evaluation sets.** Splits, example counts, and scoring units used for the benchmark comparisons.

Benchmark	Split	Size	Scoring unit
VisualPuzzles	Released collection	1,168	Question accuracy
ChartQA	Test	2,500	Relaxed answer accuracy
V*	Test	191	Question accuracy
BLINK-Jigsaw	Validation, Jigsaw	150	Question accuracy
MMVP	Full	300	Question accuracy
SAT	Real-image test	150	Question accuracy
CV-Bench	Test	2,638	Mean of 2D and 3D scores
MathVista	Testmini	1,000	Answer accuracy
MathVision	Test	3,040	Answer accuracy
VisuLogic	Original benchmark	1,000	Question accuracy
EMMA	Test, four subjects	2,788	Question accuracy

VLM training. We train on Zebra-CoT (Li et al., 2025a), using its interleaved text and image steps as teacher supervision. We use AdamW with a peak learning rate of 10^{-5} , cosine decay, a global batch size of 64, and one epoch of training, informed by prior multimodal fine-tuning recipes (Li et al., 2025a; Hu et al., 2026). We use bf16, a maximum sequence length of 8,192, 3% warmup, weight decay of 0.01, and gradient clipping at 1.0. Each backbone retains its native image processor. The unified encoder and shared reasoning backbone are optimized jointly; teacher traces are omitted at evaluation.

Thought tokens and latent sampling. We use four 512-dimensional thought tokens per block, with learned projections to and from the backbone width, and loss weights $\lambda_y = 1$ and $\lambda = 5$, taking the token budget and weights from a latent-diffusion reference configuration (Kang et al., 2025). Training retains the annotated reasoning-step sequence; the inference budget is four blocks, or 16 thought tokens, followed by greedy answer decoding with a 512-token limit for VLMs. The VLA reasoning horizon is four steps.

For flow matching, we draw $\epsilon \sim \mathcal{N}(0, I)$ and $t \sim \mathcal{U}[0, 1]$, set $z_i^t = (1-t)\epsilon + tz_i^*$, and use $v_i^* = z_i^* - \epsilon$ as the velocity target (Lipman et al., 2023). At inference, we integrate from $t = 0$ to 1 per block with one sampled trajectory and no classifier-free guidance.

RLBench. We evaluate ten tasks with a Franka Panda in CoppeliaSim, using 100 waypoint-generated training demonstrations per task and keyframe sampling (Liu et al., 2026). Inputs include a front-view RGB image resized to 384×384 , a point cloud sampled to 1,024 points, the task instruction, and proprioception. We evaluate the final checkpoint with 20 rollouts per task and seed, giving 600 rollouts across three seeds.

LIBERO. We train a separate policy for each of the four suites, each containing ten tasks and 500 training demonstrations. Inputs use third-person and wrist RGB views resized to 384×384 . We omit the point-cloud teacher because this modality is unavailable, and sample intermediate teacher steps every eight frames (Liu et al., 2026). We evaluate the final checkpoint with 50 rollouts per task and seed, giving 6,000 rollouts across the four suites and three seeds. Both benchmarks use their native success conditions.

Seeds and aggregation. Our experiments use three random seeds. For question-level accuracy or rollout success on N examples per seed, the mean is $\bar{s} = \frac{100}{3N} \sum_{r=1}^3 \sum_{j=1}^N c_{rj}$, where $c_{rj} \in \{0, 1\}$ indicates correctness or success. Grouped metrics retain the benchmark’s aggregation rule: CV-Bench averages its 2D and 3D scores. For MMVP, we report question-level accuracy over 300 questions, matching the VLMEvalKit protocol used by the relevant comparison methods (Gu et al., 2025; Li et al., 2026b). Each answer is scored independently against its correct option; the reported result is the mean of the three seed-level accuracies. Benchmark averages give equal weight to each reported benchmark or suite. Relative gains are $100(s_{\text{ours}}/s_{\text{baseline}} - 1)$; rounding follows aggregation. RLBench Table 2 retains the baseline paper’s variance convention (Liu et al., 2026). Reported gains describe differences in mean scores; they do not denote statistical significance.

VLA training. We train for 300 epochs with AdamW, using a peak learning rate of 10^{-4} , cosine decay to zero, no weight decay or warmup, and gradient clipping at 1.0 (Liu et al., 2026). Training uses eight NVIDIA A800 GPUs with bf16 and DeepSpeed ZeRO-1. The batch size is eight per device and 64 globally, without gradient accumulation. Actions are seven-dimensional, with ten flow-matching steps in the action head and a four-step reasoning horizon. We mix reasoning-to-action ratios of 1:1, 1:2, and 1:4 during training and use 1:4 at evaluation. The latent diffusion objective is defined in Section 3.

Inference speed. All inference rates in Table 2, including those of the baselines, are measured on the same single NVIDIA RTX 4090, without action chunking, using one reasoning update per four action decisions (Liu et al., 2026). We use batch size one and bf16, with 100 warmup decisions followed by 1,000 timed decisions per seed. Wall-clock timing is synchronized with CUDA before and after the timed sequence and includes observation preprocessing, host-to-device transfer, reasoning updates, and action prediction; simulator stepping and rendering are excluded. Each method uses the same inputs and timing boundary. Throughput is the number of decisions divided by elapsed time, averaged across three seeds. The reported 16.5 Hz corresponds to 60.6 ms per decision.

B.2 ABLATION PROTOCOLS

Modality grouping. Figure 3 varies how many teacher modalities share one latent space. Every variant encodes each modality with the same learned encoder, which reuses the reasoning backbone’s weights (Section 3.4); what changes is only the latent interface—the query tokens q and output projection—that defines each space. In the separate variant, every modality has its own query tokens and projection, so each is encoded into a private latent target space. For each RL-Bench pairwise variant, the named pair shares one set of query tokens and one projection, forming a common space, while the remaining modality keeps its own; full sharing maps all three modalities into a single shared space. The visual-reasoning comparison likewise gives text and image either separate spaces or one shared space. All variants retain every teacher modality, the reasoning backbone, step order, latent-token budget, and training schedule. Because the learned encoder and its capacity are identical across variants, this study isolates the effect of unifying the latent interface rather than of introducing learned compression.

Grounding objective. In Figure 4, reconstruction predicts the source teacher features from each thought block; downstream grounding predicts later teacher steps. Both variants retain the same final-output loss $\lambda_y \ell(\hat{y}, y)$ and propagate its gradients through the thought tokens into the unified encoder and shared backbone. Text CE is averaged over supervised tokens, and continuous L2 loss is averaged over target elements, using the same reduction and loss weights in both variants. Teacher targets remain fixed. Thus, the comparison changes intermediate supervision while preserving the encoder’s access to final-task supervision.

Thought-token prediction. Figure 6 uses fixed clean thought-token targets for all three objectives. L2 and cosine regression directly predict the next clean thought block. Flow matching predicts the velocity along the noisy-to-clean path, using the sampler specified in Appendix B.1. Each method conditions on the same input and preceding thought blocks and supplies its predicted tokens to the downstream reasoner. This comparison reports downstream performance for the respective prediction procedures.

B.3 ADDITIONAL RESULTS

Reasoning across backbones. We evaluate Uni-LaDiR with BAGEL-7B, Qwen3-VL-8B, and Janus-Pro at 1.5B and 7B to test whether the gains extend beyond the main backbone (Table 5). On the seven vision-centric benchmarks, the average improves over the corresponding base models by **29.5%**, **12.4%**, **35.9%**, and **51.5%**, respectively. With BAGEL, Uni-LaDiR also improves the average over Visual-OPSD by **2.5%**, leading on six of seven benchmarks; with Qwen3-VL, it improves over CoLT by **13.3%**. On BAGEL’s mathematical and logical benchmarks, the average gains are **27.9%** over the base model and **20.0%** over Uni-CoT, including a **28.3%** gain on EMMA over Uni-CoT. These improvements across model families and sizes suggest that shared latent reasoning is useful beyond a single backbone.

Table 5: **Reasoning across backbones.** Accuracy (%) on (a) seven vision-centric benchmarks and (b) four mathematical and logical benchmarks. Bold marks the best result within each backbone.

(a) Vision-centric reasoning								
Method	VisPuz.	ChartQA	V*	BLINK-J	MMVP	SAT	CV-B	Avg.
BAGEL-7B								
Base	35.00	61.82	55.49	67.33	70.33	44.67	76.03	58.67
Visual-OPSD	86.00	78.79	64.92	77.33	77.33	54.00	80.64	74.14
ThinkMorph	79.00	78.10	67.02	72.00	80.33	52.67	80.82	72.85
Uni-LaDiR	75.09	83.76	69.11	80.00	83.00	54.67	86.35	76.00
Qwen3-VL-8B								
Base	37.00	82.55	84.29	68.66	77.66	54.00	85.63	69.97
CoLT	39.04	74.70	82.20	70.00	78.00	58.00	84.00	69.42
Uni-LaDiR	60.27	86.12	85.86	78.00	84.00	67.33	89.08	78.67
Janus-Pro-1.5B								
Base	28.00	52.00	41.88	48.00	58.00	24.00	63.99	45.12
Uni-LaDiR	58.73	68.24	57.07	64.00	69.33	38.67	73.12	61.31
Janus-Pro-7B								
Base	33.50	43.08	38.22	50.67	63.33	22.00	67.83	45.52
Uni-LaDiR	69.78	75.16	64.40	70.00	76.33	46.67	80.25	68.94
(b) Mathematical and logical reasoning: BAGEL-7B								
Method	MathVista	MathVision	VisuLogic	EMMA	Avg.			
Base	72.50	35.99	28.90	28.70	41.52			
Uni-CoT	73.30	48.49	25.00	30.20	44.25			
Bagel-Zebra-CoT	72.10	43.49	0.00	20.60	34.05			
ThinkMorph	67.80	41.51	6.50	22.40	34.55			
Uni-LaDiR	79.80	57.93	35.90	38.74	53.09			

Qualitative manipulation comparison. Figure 8 compares a LaST₀ rollout with a successful Uni-LaDiR rollout from the same initial scene. LaST₀ approaches the phone but does not secure a grasp within ten decisions. Uni-LaDiR grasps the phone, transfers it to the base, and releases it, reaching success at 7.05 seconds of simulation time. This selected episode illustrates a difference at the grasping stage; aggregate success rates are reported in Table 2.

Phone placement

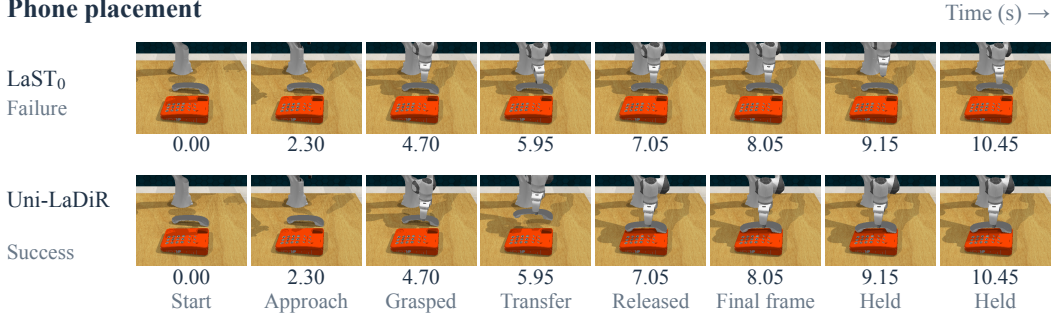


Figure 8: **Phone placement in RL Bench.** LaST₀ does not grasp the phone within ten recorded decisions (top), while Uni-LaDiR completes placement (bottom). Columns share simulation timestamps, excluding inference latency; the final frame is held after each recording ends.